

Utkarsh Pal

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SUMMARY

Computer Science post-graduate specializing in NLP and Large Language Models, with hands-on experience building AI-powered educational tools and predictive APIs.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++
- **Platforms:** Visual Studio Code, Eclipse
- **Databases:** MySQL, SQLite
- **Other Skills:** DSA, OOPs, AI, ML, Deep Learning, Computer Vision

EXPERIENCE

Java Trainer

Rajkija Engineering College
Banda, Uttar Pradesh

October 2022 – March 2023

- Conducted training sessions on core and advanced Java concepts for undergraduate students.
- Developed interactive teaching materials and provided hands-on coding exercises.

NLP/LLM Specialist

Granville Tech Pvt. Ltd.
Kampala, Uganda

February 2025 – May 2025

- Worked on real-world projects making Adaptive Learning
- Created a digital educational platform for secondary schools in Uganda that addresses high dropout rates by providing personalized, adaptive learning experiences through the integration of advanced AI technologies.

PROJECTS

Educational Content Generation API

- Developed an AI-powered API using FastAPI and transformer models to automate educational content creation, refinement, and bias detection.
- Implemented modular pipelines for high-quality text generation, readability enhancement, coherence scoring, and fairness assessment.
- Integrated Docker for seamless deployment and ensured scalability.
- The project enhances automated content generation in education and research domains.

Predictive Analysis for Manufacturing Operations

- Developed a RESTful API using Flask to predict machine downtime based on manufacturing

- data.
- Implemented Decision Tree Classifier with scikit-learn to analyze machine parameters and forecast failures.
- Designed API endpoints for data upload, model training, and real-time predictions with JSON responses.
- Processed datasets using Pandas & NumPy, achieving high model accuracy.
- Tested endpoints using Postman/cURL, ensuring robustness and efficiency.

Flappy Bird AI

- An autonomous agent designed to play the Flappy Bird game.
- Implemented using the NEAT algorithm for neuroevolution.
- Simulates gameplay environment using Python and Pygame.
- Highlights the use of evolutionary algorithms for adaptive systems.

OCR: Text extraction from images and highlights user-specified words in OCR results

- Backend developed with Python (PyTorch, Hugging Face Transformers, Qwen-VL utilities).
- Frontend designed using Gradio for interactive UI.
- Deployed on Hugging Face Spaces.

Generative AI: Multimodal text-to-image and image-to-text generation

- Combined models like Stable Diffusion, GPT-2/3, and VisionEncoderDecoderModel.
- Implemented using Python with libraries such as Transformers, Diffusers, Torch, and OpenCV.

EDUCATION

Master of Computer Applications

Banaras Hindu University
Varanasi

2023 – 2025

Bachelor of Science (Hons.) in Computer Science

Acharya Narendra Dev College, University of Delhi
New Delhi

2018 – 2021

CERTIFICATIONS AND TRAINING

Introduction to Large Language Models (LLMs)

NPTEL

Deep Learning for Computer Vision

NPTEL

Neural Networks and Deep Learning

Coursera

Supervised Machine Learning: Regression and Classification

Coursera